Amirhosein Farahmand

Tehran, Iran | amirhosein.farahmand.piruz@gmail.com | LinkedIn | GitHub | Google Scholar

Education

B.Sc. in Computer Science 2021 – 2025
University of Tehran *Tehran, Iran*

- Last Two Years GPA: **18.6**/20 (WES: **3.85**/4) | Overall GPA: **17.74**/20 (WES: **3.62**/4).

Experience

Research Assistant — Clinical Prediction Modelling and Survival Analysis Mar. 2025 – Present
Stanford University, University of British Columbia, University of Tehran *Palo Alto, California (Remote)*

- Developing and evaluating methods for estimating **time-dependent net benefit** for survival risk prediction models under **informative censoring**.
- Extended frameworks to time-dependent and competing-risk outcomes, contributing directly to the `dcurves` R package for robust net-benefit estimation.
- Secured an abstract acceptance for an **oral presentation** at the Society for Medical Decision Making 2026 Annual Meeting.

Research Assistant — Multimodal AI and Large Language Models Jun. 2024 – May 2025
GISMA University of Applied Sciences *Potsdam, Germany (Remote)*

- Designed a **multimodal Retrieval-Augmented Generation (RAG)** system using LLMs for claim verification across text and image inputs.
- Developed scalable pipelines for evidence retrieval, multimodal analysis, and claim validation against domain-specific knowledge bases.
- Co-authored research on fact-checking sustainability objectives, accepted at the **IEEE International Conference on Data Mining 2025**.

Teaching Assistant — Computer Science, Statistics, and Machine Learning Sep. 2023 – Present
University of Tehran *Tehran, Iran*

- Designed assignments and delivered instructional sessions to 250+ students across courses including **Machine Learning, Stochastic Processes, Nonlinear Optimization**, and **Probability**.
- Prepared over 25 assignment sets and evaluated 1,000+ quizzes, projects, and examinations.

Publications

Robust Decision Curve Analysis for Survival Risk Prediction Models in the Presence of Informative Censoring 2025 – Present

- Accepted for **oral presentation** at the Society for Medical Decision Making 2026 Annual Meeting, Oslo. *(Abstract, Code)*
- Authors: **Amirhosein Farahmand**, Mohsen Sadatsafavi, Abdollah Safari, and Tae Yoon Lee.

Fact-Checking Sustainability Objectives Using Multimodal Retrieval-Augmented Generation 2025

- Accepted at the **IEEE International Conference on Data Mining 2025**. *(Paper, Code)*
- Authors: Mohammad Mahdavi, **Amirhosein Farahmand**, and Abolfazl Nadi.

A Q-Learning Approach to Maintenance Policy Optimization for Series Systems 2025

- Accepted at the **11th Seminar on Reliability Theory and Its Applications**, University of Isfahan. *(Proceeding, Code)*
- Authors: **Amirhosein Farahmand**, Fatemeh Fartash, Mohammadreza Yadollahi, and Sina Zarandi.

Selected Projects

Deep Generative Models for Image Modelling *(GitHub)* 2025

- Implemented deep generative architectures including **NADE, VAE, β-VAE, GANs, Energy-Based Models**, and **NICE normalizing flows**.
- Developed **DDPM** and **DDIM** diffusion models, utilizing distillation methods for faster image-generation sampling.

Gaussian Mixture Model for Stress-Level Classification *(GitHub)* Jan. 2025

- Built a Bayesian GMM classifying stress levels via wearable sensor data, employing the Expectation-Maximization algorithm for parameter estimation.

Relevant Coursework

- Generative Models (A+)
- Non-linear Optimization (A+, Top Score)
- Artificial Intelligence (A+)
- Stochastic Processes (A+, Top Score)
- Linear Algebra (A+, Top Score)
- Scientific Computing (A+, Top Score)

Research Interests

- Machine Learning
- Survival Analysis
- Causal Inference
- AI for Healthcare

Skills

- **Programming Languages**: Python, R, C/C++
- **Frameworks/Libraries**: PyTorch, TensorFlow, Scikit-learn, NumPy, Pandas, caret, ggplot2, tidyr, dplyr
- **Tools**: Git, GitHub/GitLab, Linux, LaTeX
- **Languages**: English (IELTS Overall Band 7), Farsi (Native)